

Grade 8
Unit 4
Lesson 8 Practice

Name _____
Date _____
Period _____

For questions 1-6, simplify each expression applying the laws of exponents.

1. $\frac{n^{12}}{n^4}$

2. $\frac{28v^{10}}{-4v^7}$

3. $\frac{2b^9c^{11}}{7bc^7}$

4. $\left(\frac{3s}{-4}\right)^2$

5. $\frac{(2x^4y)^5}{-16xy^6}$

6. $\frac{(-3x^5y^3)^4 \cdot x^9y}{-9x^7y^2}$

Melinda simplified the expression $\frac{(4a^8b^2)^3}{-8a^{12}b^3}$. Her equivalent expression after simplifying using the laws of exponents was $-8a^{12}b^3$.

7. Explain whether or not you agree with her expression.

8. Find and fix her mistake.

Students in Ms. Vale's 8th grade math class, had to simplify the algebraic expressions for homework. Each student made a mistake. Circle the students' errors and show your work to justify your answer.

9. Stefan's Work	10. Monique's Work
$\frac{(4r^4y)^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{16r^6y^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{8r^{11}y^5}{2r^9y^2}$ $4r^2y^3$	$\frac{(4r^4y)^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{16r^8y^2 \cdot r^5y^3}{2r^9y^2}$ $\frac{8r^{40}y^6}{2r^9y^2}$ $4r^{31}y^4$
Correct Work:	Correct Work: